

Aditya Mitra

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Software engineer with 3 years at TCS and a recent Master's from Northeastern University. Builds full-stack web applications and RESTful APIs with FastAPI, Flask, React, and TypeScript, backed by SQL and NoSQL databases, with dashboards, logging and monitoring, and automation that reduces manual work.

EDUCATION

Northeastern University

Master of Science in Information Systems

Boston, MA

Sep. 2024 – May 2026

Amity University

Bachelor of Technology in Computer Science Engineering

Mumbai, India

Aug. 2017 – May 2021

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, PL/SQL, Bash, Shell, HTML, CSS

Backend & Frameworks: FastAPI, Flask, React, REST APIs, JWT Authentication, Node.js, Pydantic, LiteLLM, Gradio

Databases & Infrastructure: Oracle, SQLite, MongoDB, Qdrant, ELK Stack, IBM MQ, Oracle WebLogic

Cloud & DevOps: Docker, Git, GitHub, AWS, GCP, CI/CD, Linux, Hugging Face Spaces, Shell/Bash Automation

AI & ML: PyTorch, Hugging Face, LangChain, CrewAI, RAG, LoRA/PEFT, Vector Search, Tesseract OCR, ROUGE

Fundamentals: Data Structures & Algorithms, OOP, System Design, Concurrency, Production Reliability, Technical Documentation, Agile

EXPERIENCE

Systems Engineer (Promoted)

Oct. 2023 – Aug. 2024

Tata Consultancy Services

Mumbai, India

- Led a team of 3 to roll out a full ELK Stack pipeline (Elasticsearch, Logstash, Kibana) for core banking systems, giving engineers production visibility and making deployments safer.
- Automated roughly 60% of manual infrastructure work with Python and shell scripts, cutting human error and speeding up incident response.
- Partnered with engineering and compliance teams on system design and prioritization, driving root-cause fixes over symptom patching.

Assistant Systems Engineer

Oct. 2021 – Sep. 2023

Tata Consultancy Services

Mumbai, India

- Built an internal React and JavaScript monitoring dashboard consolidating server health, alerts, and logs into one view, accelerating response to production issues.
- Tuned PL/SQL scripts to cut report generation time by 60%, and wrote shell scripts to push monitoring updates and system alerts across the production pipeline.
- Managed IBM MQ and Oracle WebLogic infrastructure for core banking transactions, and automated security audits and log investigations on Red Hat Linux.
- Earned Star of the Month and Best Team awards for production reliability and automation impact across multiple core banking releases.

PROJECTS

AlgoSensei: GenAI DSA Tutor with RAG and Leakage Gate | *Python, FastAPI, REST, Qdrant, Docker*

- Built a Socratic tutoring app covering 175 LeetCode problems across 14 DSA patterns, with a FastAPI backend, RAG retrieval over Qdrant, JWT auth, and SQLite for user data and progress.
- Designed a two-stage leakage gate (regex and semantic checks) that achieved 0% solution leakage in testing, with up to three regeneration retries on failure.
- Added a Tesseract OCR pipeline turning LeetCode screenshots into tutoring sessions, and a provider-agnostic LLM client (OpenRouter / OpenAI), deployed via Docker on Hugging Face Spaces.

Multi-Agent Personal Productivity Assistant | *Python, CrewAI, Flask, Pydantic*

- Built a 6-agent CrewAI pipeline (Task Manager, Researcher, Scheduler, Analytics, Career Coach, Controller), each agent with its own tools and Pydantic-typed input/output contracts.
- Diagnosed a CrewAI context-window overflow and resolved it with explicit per-task context isolation, keeping the pipeline reliable as task lists grew.
- Built a Flask backend with real-time SSE streaming and SQLite persistence, and wrote 5 custom Python tools.

Medical Question Answering with Fine-Tuned Llama-3 | *Python, PyTorch, LoRA/PEFT, Hugging Face*

- Fine-tuned Llama-3.2-1B-Instruct on MedQuAD with LoRA/PEFT (Kaggle T4, 500 steps, FP16, cosine warmup).
- Evaluated against the base model with ROUGE, analyzing failure modes on rare and ambiguous queries.
- Deployed real-time inference through a Gradio interface with an automatic medical disclaimer for responsible use.